Evolutionary and Institutional Economics Review (2018) 15:477–493
https://doi.org/10.1007/s40844-018-0111-z

ARTICLE

Identification of conduit jurisdictions and community structures in the withholding tax network

Tembo Nakamoto[1] · **Yuichi Ikeda**[1]

Published online: 1 November 2018
© Japan Association for Evolutionary Economics 2018

Abstract

Due to economic globalization, international tax avoidance has emerged as a well-known global issue. To contribute to providing the solution of international tax avoidance, we tried to investigate which part of the network is vulnerable. Specifically, focusing on treaty shopping, which is one of the international tax avoidance schemes, we attempt to find which jurisdictions are likely to be used for treaty shopping from the viewpoint of tax rates and reveal the relationships between jurisdictions used for treaty shopping and the others. For that purpose, based on withholding tax rates imposed on dividends, interest, and royalties, we produced the withholding tax network expressed as weighted graphs, computed the centralities and detected the communities. As a result, we identified the jurisdictions used for treaty shopping and pointed out that there are community structures. The results of our study suggested that fewer jurisdictions need to introduce more regulations for the prevention of treaty shopping worldwide.

Keywords International taxation · International tax avoidance · Network analysis

JEL Classification H26 · K34 · F69

✉ Tembo Nakamoto
nakamoto.tembo.75w@st.kyoto-u.ac.jp

Yuichi Ikeda
ikeda.yuichi.2w@kyoto-u.ac.jp

[1] Graduate School of Advanced Integrated Studies in Human Survivability, 1, Yoshida-Nakaadachi-cho, Sakyo-ku, Kyoto 606-8306, Japan

1 Introduction

Economic globalization has great influences on multinationals' business activities and international taxation. The activities of multinationals cross more borders and become more complex. Accordingly, the number of tax treaties regulating conflicts of each jurisdiction's[1] right of taxation increases from 1,200 in 1975 to 3200 today worldwide. Now each jurisdiction's tax laws and treaties overlap and function as a single legal system (Ogata 2016) or network.

However, it occurs a severe problem. When tax laws and treaties of multiple jurisdictions overlap, the differences among them create opportunities for unexpected reductions in the tax burden because each jurisdiction often legislates their laws and treaties individually and without consideration of the overlapping. The phenomenon, that is commonly called international tax avoidance, is one of the side effects of economic globalization.

Meanwhile, in recent years, network science has made remarkable progress by offering accurate network information in various fields, and found networks follow the same principle, although the networks come from different fields (Barabsi 2016). They have revealed interesting results that had hitherto not been found in a broad realm from sociology to physics.

Our study focuses on "Treaty Shopping," which is one of international tax avoidance schemes, and attempts to highlight interesting aspects that have not yet been found far in the realm of international taxation. We make use of the methods and knowledge of network science because international tax avoidance emerges through tax laws and treaties' working as one network.

The previous studies incorporating the method of network science only studied withholding tax on dividends (Polak 2014; Van't Riet and Lejour 2014; Hong 2017), but the other passive incomes, that are interest and royalties, are also important, and relationships among each jurisdiction have not been analyzed in a context of treaty shopping, even though experts pointed out the existence (Honjo 2012). "Treaty Shopping" is clarified through making these points clear by the methods of network science.

2 International tax avoidance

2.1 International taxation system

Companies are subject to various forms of taxation in the process of conducting cross-border transactions, most importantly corporate tax and withholding tax. According to international custom, each jurisdiction has the primary right to tax income which arises in or is derived from that jurisdiction (Arnold 2016). Therefore, each jurisdiction imposes corporate tax on business income arising in its jurisdiction

[1] The meaning of "jurisdiction" is almost the same as that of "country and region."

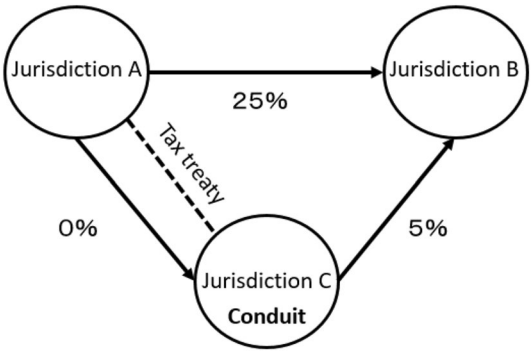


Fig. 1 Treaty shopping

and imposes withholding tax on dividends, interest, and royalties derived from its jurisdiction and paid to other jurisdictions. Even if dividends, interest and royalties are paid to companies in the same corporate group, withholding tax is imposed as long as the group company is in another jurisdiction.

However, when dividends, interest, and royalties are paid to jurisdictions that conclude a tax treaty, the amount of withholding tax is often lower than usual. This is because many tax treaties have provisions to reduce the withholding tax rate or exempt the company from paying the tax altogether. This reduction or exemption benefit is generally called tax treaty benefit. The purpose of offering a tax treaty benefit is to encourage economic activities between the two contracting jurisdictions, so the reduction or exemption is applied only to dividends, interest, and royalties paid to the other contracting state.

2.2 Treaty shopping

The tax treaty benefits considered the amount of withholding tax that may be reduced by circumventing the third jurisdiction rather than paying dividends, interest, and royalties directly to other jurisdictions. Figure 1 shows an example. Suppose that jurisdiction A (a source jurisdiction) imposes a 25% withholding tax, jurisdiction C imposes a 5% withholding tax and the withholding tax is waived for dividends, interest and royalties paid from jurisdiction A to jurisdiction C because jurisdiction A has concluded a tax treaty with jurisdiction C. In this case, if a company located in jurisdiction A pays dividends, interest, and royalties directly to a company located in jurisdiction B, a 25% withholding tax is imposed. However, if a company located in jurisdiction A pays its dividends, interest, and royalties to a company located in jurisdiction B via jurisdiction C (a conduit jurisdictions), it is exempted from the withholding tax for payments made between jurisdiction A and jurisdiction C because of the A=C tax treaty, and jurisdiction C imposes a 5% withholding tax on payments made between jurisdiction C and jurisdiction B. In a case where a company located in jurisdiction A pays its dividends, interest, and royalties to a company located in jurisdiction B, if it pays them via jurisdiction C, it can avoid a 20% withholding tax, compared to paying them directly.

Reducing the amount of tax through the unexpected use of tax treaty benefits described above is called "Treaty Shopping." Treaty shopping is not illegal unless violates the provisions of a jurisdiction's tax law or tax treaty, but particularly if it is done solely to reduce the amount of tax, it is called "treaty abuse" (Marian 2016) or "improper use of the convention" (OECD 2015a) and should be corrected.

For companies engaged in treaty shopping, this is just one of the ways to avoid tax. However, for source jurisdictions, this is one of the ways in which their rights to impose taxation are eroded by third jurisdictions (conduit jurisdictions). It is source jurisdictions that originally have the right to impose taxation since the dividends, interest, and royalties are income derived from their jurisdiction. Nevertheless, it is the third jurisdictions that impose their tax on these incomes. In this way, it has been found that using a particular bilateral tax treaty erodes the tax sources of jurisdictions that did not conclude the bilateral tax treaty. In other words, vulnerabilities in one jurisdiction's tax treaty potentially erode other jurisdictions' tax bases. This is why treaty shopping is regarded as a form of tax base erosion and is viewed as a problem in the area of international taxation.

3 Withholding tax network

3.1 Datasets

The withholding tax rate data relating to dividends, interest, and royalties come from Ernst and Young (2017). We analyzed 165 jurisdictions, which include important jurisdictions of each area in the data. With respect to the withholding tax rate on dividends, many tax treaties apply different tax rates depending on the percentage of holdings. Generally, the higher the shareholding ratio, the lower the tax rate. The interest in this study is in the tax burden that arises when the profit is transferred within a corporate group. This is because international tax avoidance by multinationals is often done through the transfer of their profits between group companies. Therefore, assuming that the dividend was generated by a 100% wholly owned subsidiary, the lowest tax rate is applied. Regarding interest, some jurisdictions have applied different tax rates for non-bank deposits and bank deposits. For the same reason, it is assumed that the tax rate for the non-bank deposit is applied.

Some tax treaties have other requirements for granting withholding tax exemptions or reductions, apart from the shareholding ratio mentioned above. We consider that all requirements are met in this case. Moreover, the applied withholding tax rate is not always clear because the wording in a given tax treaty and the relationship between domestic tax laws and tax treaties are not always clear. In these cases, the lowest tax rate among the possible withholding tax rates is applied. On the other hand, although some jurisdictions do not impose withholding tax according to their domestic tax laws, their tax treaties determine the withholding tax rate. Since the primary purpose of tax treaties is usually to avoid international double taxation, it is considered that withholding tax is not imposed, but only in this case, we assume that the tax rate stipulated in tax treaties is to be applied.

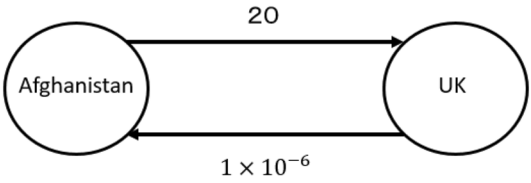


Fig. 2 An example of Afghanistan and the United Kingdom in the weighted multi-directed graph

It is also necessary to consider corporate taxes in the third jurisdiction when calculating the strict amount of tax. This is especially true in the case of dividends. However, corporate taxes in the third jurisdiction are not considered in this study because only a few jurisdictions impose corporate tax on dividend income today[2] (Dittmer 2012).

3.2 Constructing the network

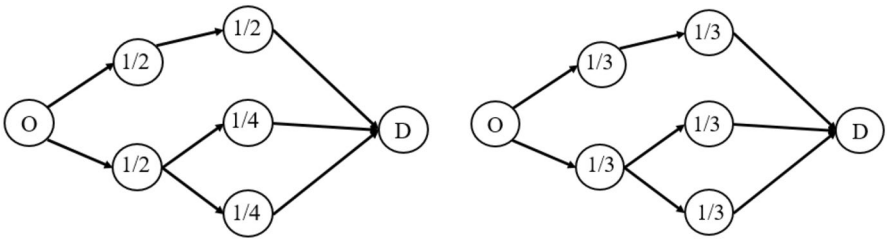
As a withholding tax network, we produce weighted multi-directed graphs. In the graph, the vertices represent jurisdictions, and all pairs of vertices are connected by arcs because, logically, any jurisdiction can pay dividends, interest, and royalties to any other jurisdictions. Every arc has the withholding tax rate imposed on dividends, interest, and royalties as a weight, respectively. Moreover, 1×10^{-6}, which is small enough not to affect the validity of our analysis, compared with other tax rates, is given to arcs whose tax rates are 0%, so that the paths include many arcs whose tax rates 0% were not counted as shortage paths. Figure 2 shows an example of Afghanistan and the United Kingdom. Afghanistan imposes a 20% withholding tax on dividends paid to the United Kingdom and the United Kingdom imposes a 0% withholding tax on dividends paid to Afghanistan. The arc from Afghanistan to the United Kingdom B has 20 as the weight, while the arc from the United Kingdom to Afghanistan has 1×10^{-6}.

4 Methods of network analysis

4.1 Centrality calculation

We try to estimate which jurisdiction is important for treaty shopping from the viewpoint of withholding tax rates by calculating centrality. Centrality is an index that shows how important a vertex is for the overall network. The higher the value, the more important the vertex is in the network. In this study, centrality is calculated based on the idea that the vertex contributing to the shortest paths, that is the minimum weighted paths, in the network is essential for the whole network because jurisdictions used for treaty shopping are located on the shortest paths.

[2] Many countries have introduced participation exemptions. For example, in Japan, corporate tax is exempted for 95% of dividend income.



(a) Load centrality. **(b)** Betweenness centrality.

Fig. 3 Difference in the treatment of multiple shortest paths between load centrality and betweenness centrality. O indicates origin and D indicates destination. **a** Load centrality. **b** Betweenness centrality

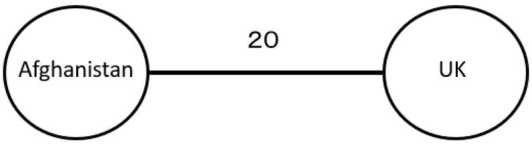


Fig. 4 An example of Afghanistan and the United Kingdom in the weighted undirected graph, in the case of no threshold or threshold 30–20%

We use load centrality as the centrality of the graph (Goh et al. 2001). Load centrality was originally designed to calculate how much data processing capacity each passing point requires for the efficient movement of data packets in a network such as the Internet. The definition is as follows: Suppose one data packet is sent from vertex i to vertex j. The data packet moves along the minimum weight path. When one data packet is sent among all vertices like this, the amount of the data packets which passed through vertex k is represented by l_k. The load centrality of vertex k is given by

$$C_l(k) = \sum_{i \neq j \neq k} l_{i,j}(k) \tag{1}$$

Even though load centrality is sometimes thought to be similar in concept to betweenness centrality (Freeman 1979), the results differ when a graph has multiple shortest paths (Brandes 2008). This is because load centrality assumes a data packet is divided equally at the fork of shortest paths, while betweenness centrality is divided equally by the number of shortest paths (Fig. 3). In this study, we think of data packets as the amount of dividends, interest, and royalties and withholding tax rates as weight to find the vulnerabilities of the worldwide legal system (Sect. 1).

4.2 Community detection

For the purpose of clarifying relationships between jurisdictions using treaty shopping and jurisdictions used for treaty shopping, we try to detect communities in the withholding tax network. When detecting community, the weighted multi-directed graphs were transformed into weighted undirected graphs because the way of community detection described below can be applied only

Table 1 Ranking of dividends' centrality

Rank	Jurisdiction	Centrality	Rank	Jurisdiction	Centrality
1	UK	0.070639	18	Malta	0.015731
2	UAE	0.058244	19	USA	0.01268
3	Kuwait	0.044943	20	South Africa	0.011743
4	Netherlands	0.027664	21	Mexico	0.010967
5	Cyprus	0.024464	22	Oman	0.010475
6	Hong Kong	0.0239	23	Denmark	0.010308
7	Singapore	0.0228	24	Hungary	0.010111
8	Switzerland	0.022725	25	Belgium	0.009487
9	Mauritius	0.022512	26	France	0.009139
10	Spain	0.02209	27	Lithuania	0.007719
11	Luxembourg	0.020043	28	Georgia	0.007616
12	Lucia	0.018178	29	Bulgaria	0.006243
13	Bahrain	0.018153	30	Colombia	0.006188
14	Malaysia	0.018115	31	Zambia	0.006114
15	Qatar	0.017946	32	Tunisia	0.006075
16	Ireland	0.017523	33	Germany	0.005091
17	Estonia	0.016039			

to undirected graphs. When withholding tax rates differ between the same two jurisdictions, the higher withholding tax rate is put on edges as a weight in the weighted undirected graphs because the higher rate is suitable for measuring strength of the relationships. Since the higher the withholding tax rate of the edge, the lower is the possibility of the edges being used for treaty shopping, edges of the weighted undirected graphs were removed according to thresholds with the intention of searching at which threshold value a community structure can be found. The threshold was provided every 5% in the range of 30–0%, that is, 30%, 25%, 20%, 15%, 10%, 5%, and 0%. This is because few jurisdictions impose a withholding tax higher than 35%. Figure 4 shows an example of Afghanistan and the United Kingdom. In the weighted undirected graph of dividends, 20 is put on the edge between Afghanistan and the United Kingdom as a weight because Afghanistan imposes a 20% withholding tax on the dividends paid for British company, although no withholding tax is imposed on the dividends paid for Afghan company (Fig. 2). The edge between Afghanistan and the United Kingdom is removed when the threshold is 15%, 10%, 5%, or 0%.

Then, the way of community detection was applied to each of the weighted undirected graphs removing their edges depending on the given thresholds. Communities whose modularities are highest among all threshold values are chosen finally. In addition, correlation coefficients are calculated between the centralities in the weighted multi-directed graphs (Tables 1, 2, 3) and the centralities in the weighted undirected graphs with each threshold value in order to check that the result of the centralities is not affected too much by providing the threshold.

Table 2 Ranking of interest's centrality

Rank	Jurisdiction	Centrality	Rank	Jurisdiction	Centrality
1	UAE	0.041796	20	Cyprus	0.009644
2	Switzerland	0.039489	21	Russia	0.009492
3	Germany	0.035639	22	Bahrain	0.009354
4	France	0.033565	23	Latvia	0.009251
5	UK	0.032811	24	Italy	0.008287
6	Hungary	0.025905	25	Belarus	0.006977
7	Canada	0.024145	26	Mauritius	0.006841
8	Netherlands	0.023775	27	Oman	0.006553
9	Sweden	0.023619	28	Ukraine	0.006507
10	Luxembourg	0.022488	29	Bulgaria	0.006357
11	Kuwait	0.019532	30	Israel	0.006243
12	Norway	0.016801	31	Algeria	0.006228
13	Ireland	0.016736	32	Japan	0.006183
14	Czech	0.016253	33	Seychelles	0.006154
15	USA	0.014057	34	Portugal	0.006108
16	Estonia	0.013082	35	Zambia	0.006101
17	Malta	0.011142	36	Cameroon	0.006067
18	Austria	0.01061	37	Denmark	0.005954
19	South Africa	0.010495			

Table 3 Ranking of royalties' centrality

Rank	Jurisdiction	Centrality	Rank	Jurisdiction	Centrality
1	UAE	0.079455	20	Germany	0.016731
2	Switzerland	0.079043	21	Macau	0.01241
3	France	0.05139	22	Canada	0.012097
4	Hungary	0.048268	23	Monaco	0.01029
5	Mauritius	0.047201	24	South Korea	0.009735
6	Sweden	0.045731	25	South Africa	0.009731
7	Netherlands	0.042169	26	Japan	0.008836
8	Norway	0.037786	27	Denmark	0.007992
9	Cyprus	0.034533	28	Italy	0.007951
10	Ireland	0.029864	29	Guernsey	0.007783
11	Luxembourg	0.029044	30	Estonia	0.007717
12	Senegal	0.029034	31	Slovak	0.006905
13	UK	0.024726	32	Austria	0.006784
14	Malta	0.02437	33	Zambia	0.006121
15	Gabon	0.024012	34	Libya	0.005969
16	USA	0.019653	35	Liechtenstein	0.005768
17	Spain	0.018766	36	Belgium	0.005208
18	Bahrain	0.018091	37	Russia	0.005146
19	Latvia	0.01757			

4.2.1 Modularity

A community is a subgraph closely connected in a graph. Although there is no strict definition of "community" in network science, it is thought that the network has a community structure, if the graph can be divided into subgraphs where the edges connected has density. To evaluate only significant community structures, we use modularity, which is often used as an index of evaluating community structure (Newman and Girvan 2004; Newman 2004). Modularity is calculated as follows:

$$Q = \frac{1}{2M} \sum_{i,j}^{N} \left(a_{ij} - \frac{k_i k_j}{2M} \right) \delta[C(i), C(j)] \tag{2}$$

where M is the total number of the edges, $C(i)$ is the subset including vertex i, and $\delta[C(i), C(j)]$ is 1 if $C(i)$ and $C(j)$ are the same subset, otherwise it is 0. a_{ij} represents the weight of the edges between vertex i and vertex j, and $k_i\, k_j$ represents the expected weight of the edges assigned randomly to vertex i.

Modularity is normalized so that the maximum value is 1. Therefore, it can be said that there is a strong community structure in the network as it moves closer to 1, but from an empirical point of view, if it is larger than about 0.3, it is often considered that the network has a community structure.

4.2.2 Louvain method

Communities are detected through the Louvain method (Blondel et al. 2008). For the detection, the Louvain method optimizes local and aggregates vertices as follows: First, it assumes that every vertex belongs to every community and calculates how much the modularity rises, if each vertex belongs to the same community with an adjacent vertex:

$$\Delta Q = \left[\frac{\sum_{in} + k_{i,in}}{2M} - \left(\frac{\sum_{tot} + k_{i,in}}{2M} \right)^2 \right] - \left[\frac{\sum_{in}}{2M} - \left(\frac{\sum_{tot}}{2M} \right)^2 - \left(\frac{k_i}{2M} \right)^2 \right] \tag{3}$$

where if vertex i belongs to a community, $\sum_{in}$ represents the sum of the weights of edges in the community, $\sum_{tot}$ represents the sum of the weights of all the edges adjacent to the vertices in the community, k_i represents the sum of the weights of edges whose end point is vertex i, and $k_{i,in}$ represents the sum of the weights of edges in the community whose endpoint is vertex i. The combination of vertices with the greatest modularity rise is specified (the first step). Next, combinations of the specified vertices in the first step are aggregated into one vertex, and the size of the network is reduced (the second step). The first stage and the second stage are recursively repeated until the modularity converges. The characteristic of the Louvain method is not to assume all vertices belong to one community, but to assume each belongs to each community.

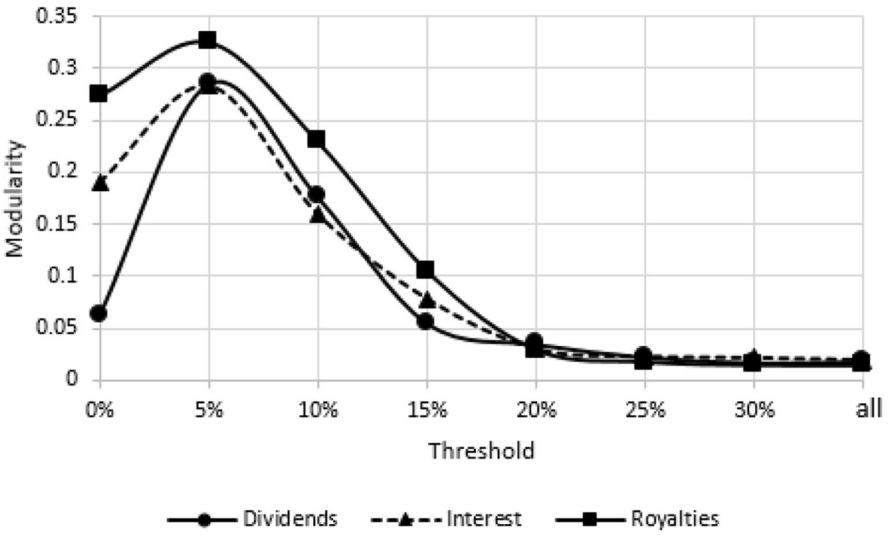


Fig. 5 Changes in modularity

5 Results

5.1 The centralities

Tables 1, 2 and 3 show rankings and jurisdictions whose centralities were higher than 0.05 about dividends, interest, and royalties without any threshold. The top jurisdictions differed markedly depending on dividends, interest, and royalties, and the centrality of interest is much lower than those of others.

In the area of international taxation, the Netherlands, Cyprus, Estonia, Hungary, Luxembourg, Malta, Switzerland, the United Kingdom and so on are commonly known to be used for treaty shopping (Diamond et al. 2017). Comparison of the jurisdictions whose centralities were higher than 0.05 in our calculation and the jurisdictions commonly known shows that our method is capable of identifying the jurisdictions used for treaty shopping effectively.

5.2 Communities founded

Figure 5 shows changes in modularity of the weighted undirected graphs according to the thresholds. When threshold is 5%, all graphs recorded the highest modularity, which is 0.2853738 (dividends), 0.283715 (interest), and 0.325377 (royalties). Because of this, we detected community as for the weighted undirected graphs with a 5% threshold. Table 4 shows correlation coefficients between the centralities in the weighted multi-directed graphs (Table 1, 3) and the centralities in the weighted undirected graphs with each threshold value. Since all coefficients

Table 4 Correlation coefficients of the centralities

Threshold (%)	Coefficient
(a) Dividends	
30	0.964963813
25	0.964979316
20	0.96515412
15	0.963629098
10	0.965430042
5	0.959738913
0	0.925707564
(b) Interest	
30	0.951327102
25	0.951316117
20	0.951212308
15	0.950376936
10	0.937609801
5	0.912283135
0	0.920637142
(c) Royalties	
30	0.944070249
25	0.943941746
20	0.943555807
15	0.936168543
10	0.93951127
5	0.911668238
0	0.880016323

Table 5 Communities (dividends)

Community 1 (61)		Community 2 (48)	Community 3 (19)
UK	Estonia	Cyprus	UAE
Kuwait	Malta	Hong Kong	Singapore
Netherlands	USA	Saint Lucia	Malaysia
Switzerland	South Africa	Bahrain	Qatar
Spain	Mexico	Mauritius	
Luxembourg	Denmark	Oman	
Ireland	Hungary		

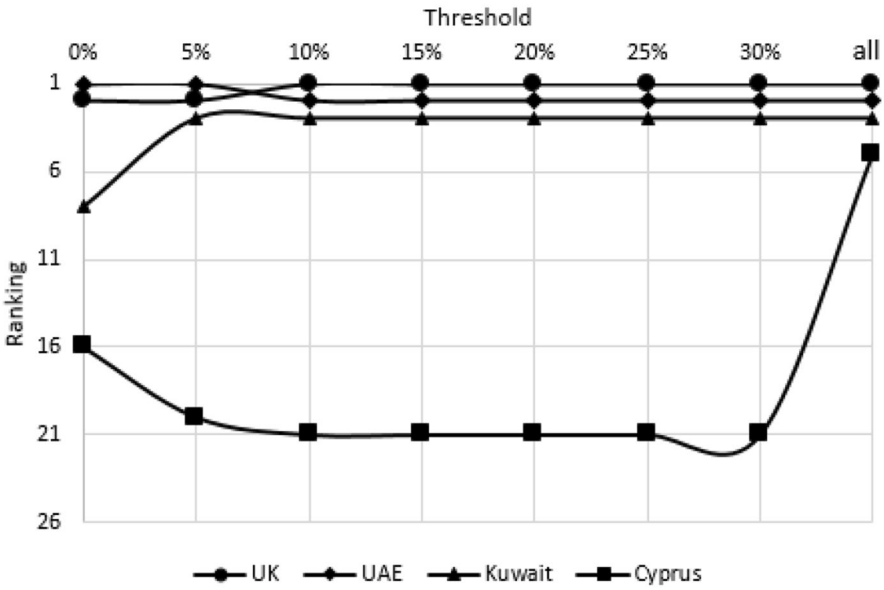


Fig. 6 Change in ranking of top 3 jurisdictions and Cyprus in the case of dividends

are around 0.9, it can be said for providing the thresholds not to influence the results of the centralities so much.

5.2.1 Dividends

The graph of dividends can be divided into four communities. Community 1 has 61 jurisdictions, community 2 has 48 jurisdictions, community 3 has 19 jurisdictions, community 4 has 3 jurisdictions. Table 5 shows which communities the jurisdictions whose centrality is higher than 0.01 in the weighted multi-directed graph belong to (Table 1). The other 34 jurisdictions are isolated. Although the change in the threshold values did not give big influences to the value of the centrality, some jurisdictions are fluctuating. Figure 6 shows an example of Cyprus. Top 3 jurisdictions indicate jurisdictions whose centralities are the highest in the weighted multi-directed graph. Here, the United Kingdom, the United Arab Emirates and Kuwait correspond to the top 3 jurisdictions (Table 4). The rank of Cyprus fell sharply after a threshold 30%, while the ranks of top 3 jurisdictions were stable.

5.2.2 Interest

Four communities are detected relating to interest. Community 1 has 55 jurisdictions, community 2 has 48 jurisdictions, community 3 has 13 jurisdictions, and community 4 has 2 jurisdictions. Table 6 shows which communities the jurisdictions whose centrality is higher than 0.01 in the weighted multi-directed graph belong to (Table 2). The other 47 jurisdictions are isolated. Many European countries

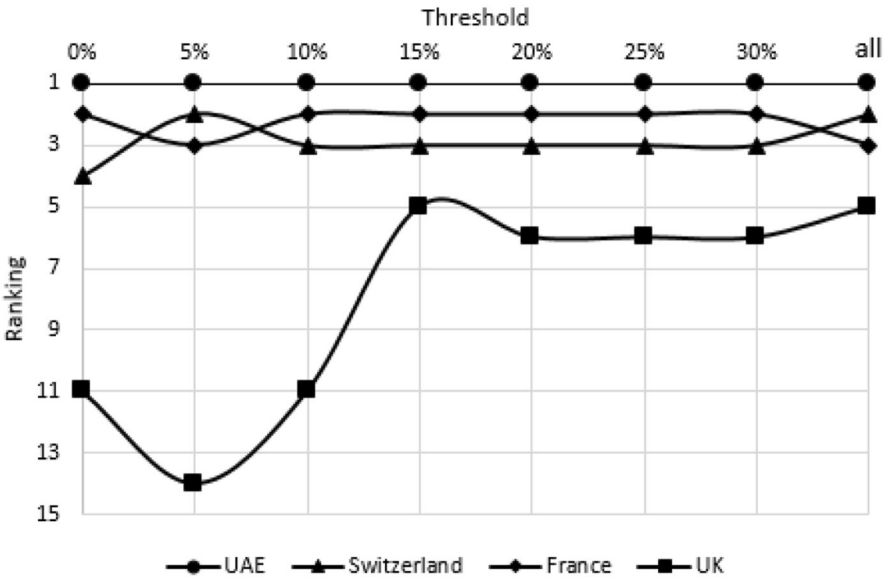


Fig. 7 Changes in ranking of top 3 jurisdictions and the United Kingdom in the case of interest

Table 6 Communities (interest)

Community 1 (55)		Community 2 (48)	Community 3 (13)
Switzerland	Czech	Hungary	UAE
Germany	Russia	Netherlands	
France	Austria	Sweden	
Kuwait	Malta	Canada	
UK		Luxembourg	
Ireland		Norway	
USA		Estonia	

belong to Community 1 or 2. This may be because the European Union (EU) has issued an "Interest Directive" (2003/49/EC), which exempts companies from paying withholding tax on interest paid within the EU member jurisdictions, to help create a single market. A few jurisdictions whose ranks were fluctuating by change in the threshold values. Figure 7 shows an example of the United Kingdom. The rank of the United Kingdom fell sharply after a threshold 10%, even though the ranks of top 3 jurisdictions were stable.

5.2.3 Royalties

We identified five communities. Community 1 has 41 jurisdictions, community 2 has 33 jurisdictions, community 3 has 35 jurisdictions, community 4 has 9

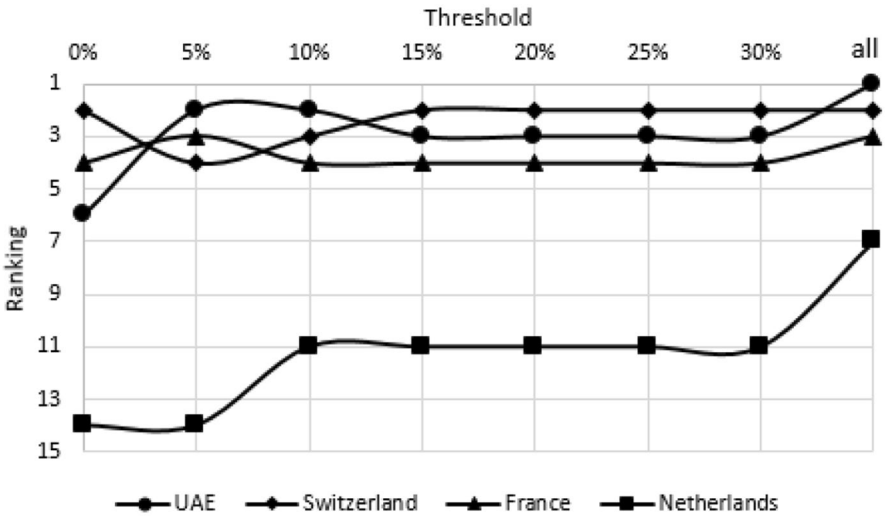


Fig. 8 Changes in ranking of top 3 jurisdictions and the Netherlands in the case of royalties

Table 7 Communities (royalties)

Community 1 (41)	Community 2 (33)	Community 3 (35)	Community 4 (9)	Community5 (4)
Switzerland	France	Senegal	UAE	Gabon
Hungary	Sweden	Cyprus	Mauritius	
Netherlands	Norway	Bahrain		
Ireland	Luxembourg	Macau		
Malta	USA	Monaco		
Spain	Latvia			
Germany	UK			
Canada				

jurisdictions, and community 5 has 4 jurisdictions. Table 7 shows which communities the jurisdictions whose centrality is higher than 0.01 in the weighted multi-directed graph belong to (Table 3). The other 43 jurisdictions are isolated. Many European countries belong to Community 1 or 2. This may be because the EU has issued a "Royalty Directive" (2003/49/EC), which exempts companies from paying withholding tax on royalties paid within the EU member jurisdictions. Figure 8 shows the rank of the Netherlands as an example of a few jurisdictions whose ranks were fluctuating by the threshold. The rank of the Netherlands fell when the threshold value was 30% and 10%.

6 Discussion

We cannot find a relationship between the detected communities and each jurisdiction's location. It might cause the United Arab Emirates not to be suitable for treaty shopping that the size of the community to which it belongs is small, although its centrality is high. The jurisdictions whose ranks vary much by change in the threshold, such as Cyprus, the United Kingdom, and the Netherlands, might be crucial with jurisdictions whose withholding tax rates are high.

To resolve the vulnerable elements of tax treaties, it is necessary to amend treaties. The realm of international taxation is trying to deal with the vulnerable arising from a single network that tax laws and treaties of each jurisdiction form by a multilateral treaty (OECD 2015b). The multilateral treaty aims to amend tax treaties of each jurisdiction uniformly, instead of sequentially revising tax treaties through bilateral negotiations as in the past. When a jurisdiction ratifies a multilateral treaty, the new provisions stipulated in the multilateral treaty overwrite the existing provisions of any tax treaties that the jurisdiction has concluded, producing the same effect as that of an amendment to the tax treaty. In order for many jurisdictions to ratify the multilateral treaty, the multilateral treaty can effectively include only the minimum standard deemed necessary for the prevention of treaty abuse. From the intention of trying to be "fair" to each jurisdiction, it seems possible only to make revisions to each jurisdiction's tax treaties "uniformly." However, according to our study, only some jurisdictions are conducive to treaty shopping, so to solve treaty shopping, it seems sufficient to amend the tax treaties of those jurisdictions for the time being. This would mean fewer jurisdictions that need to obtain consensus, which we believe would allow additional provisions to be introduced into the tax treaties[3].

7 Conclusion

Due to economic globalization, each jurisdiction's tax laws and treaties have been forced to work as a single network. However, the jurisdictions have not made their laws and treaties under the assumption that the laws and treaties function as an element of one network, so unexpected results have been brought. We thought that the results are exactly international tax avoidance. To contribute to providing the solution of international tax avoidance, we tried to investigate which part of the network is vulnerable. Specifically, focusing on treaty shopping, which is one of international tax avoidance schemes, we attempt to identify which jurisdiction are likely to be used for treaty shopping from the viewpoint of tax rates and reveal the relationships between jurisdictions used for treaty shopping and the others. For this purpose, based on withholding tax rates imposed on dividends, interest, and royalties,

[3] However, seeking prevention measures only for specific countries also has the risk of disrupting efforts towards international tax avoidance. About how the harmful tax competition taking the initiative by the OECD was abandoned, see Sharman (2006).

we produced the withholding tax network expressed as weighted graphs, computed the centralities and detected the communities.

As a result, we were able to identify jurisdictions used for treaty shopping from the viewpoint of the withholding tax rates because our results matched with the jurisdictions known to be used for treaty shopping. The community structures were founded at a 5% threshold, and all jurisdictions can be divided into four or five communities.

As a future task, we would like to study the relation between withholding tax rates and economic activities of multinationals by analyzing flows of dividend, interest, and royalties.

Acknowledgements The present study was supported by the Ministry of Education, Science, Sports, and Culture, Grants-in-Aid for Scientific Research (B), Grant No. 17KT0034 (2017-2019) and Exploratory Challenges on Post-K computer (Studies of Multi-level Spatiotemporal Simulation of Socioeconomic Phenomena).

Compliance with ethical standards

Conflict of interest statement On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

Arnold BJ (2016) International tax primer. Kluwer Law International BV, The Netherlands

Barabsi A (2016) Network science. Cambridge University Press, Cambridge

Blondel VD, Guillaume J, Lambiotte R, Lefebvre E (2008) Fast unfolding of communities in large networks. J Stat Mech: Theory Exp 2008:P10008

Brandes U (2008) On variants of shortest-path betweenness centrality and their generic computation. Social Netw 30(2):136–145

Dittmer P (2012) A global perspective on territorial taxation. Special Report No. 202, Tax Foundation

Diamond WH, Diamond DB, Byrnes William H, Munro RJ RJ (2017) Tax havens of the world. Matthew Bender & Company Inc, New York

Ernst Y (2017) Worldwide corporate tax guide 2017

Freeman LC (1979) Centrality in social networks conceptual clarification. Social Netw 1:215–239

Goh K, Kahng B, Kim D (2001) Universal behavior of load distribution in scale-free networks. Phys Rev Lett 87(27):1–4

HJI Panayi C, Avi-Yonah RS (2010) rethinking treaty-shopping: lessons for the European Union. U of Michigan Law & Econ, Empirical Legal Studies Center Paper No. 10–002

Hong S (2017) Tax treaties and foreign direct investment: a network approach. Int Tax Publ Finance 25:1277–320

Honjo T (2012) NEW TAX TIDBIT offshore sekai no hanashi (3): maboroshi no offshore sekai-sono jitsuzou ha kakusareteiru [The story of offshore world (3): phantom offshore world - the real image is hidden]. Int Tax 32(11):138–140 (in Japanese)

Marian O (2016) Unilateral responses to tax treaty abuse: a functional approach. Brooklyn J Int Law 41(3):1157–1183

Newman MEJ, Girvan M (2004) Finding and evaluating community structure in networks. Phys Rev E 69:026113

Newman MEJ (2004) Analysis of weighted networks. Phys Rev E 70:056131

OECD (2015) Preventing the granting of treaty benefits in inappropriate circumstances, Action 6—2015 Final Report, OECD/G20 Base Erosion and Profit Shifting Project. OECD Publishing, Paris

OECD (2015) Developing a multilateral instrument to modify bilateral tax treaties, Action 15—2015 Final Report, OECD/G20 Base Erosion and Profit Shifting Project, OECD Publishing, Paris

Ogata K (2016) Debate about anti-tax avoidance measures under the beps project etc. Fin Rev 126:196–225 (in Japanese)
Polak S (2014) Algorithms for the network analysis of bilateral tax treaties. Master's thesis conducted at UvA/CWI/CPB
Sharman JC (2006) Heavens in a storm: the struggle for global tax regulation. Cornell University Press, New York
Van't Riet M, Lejour (2014) A ranking the stars: network analysis of bilateral tax treaties. CPB Discussion Paper 290